Course notes

Addition and double-angle formulas

Property · Addition formulas. For all real numbers a and b:

$$\cos(a-b) = \cos a \cos b + \sin a \sin b, \qquad \cos(a+b) = \cos a \cos b - \sin a \sin b,$$

$$\sin(a-b) = \sin a \cos b - \cos a \sin b, \qquad \sin(a+b) = \sin a \cos b + \cos a \sin b.$$

Proof. In a direct orthonormal frame, let $\overrightarrow{p}\begin{pmatrix}\cos a\\ \sin a\end{pmatrix}$ and $\overrightarrow{q}\begin{pmatrix}\cos b\\ \sin b\end{pmatrix}$: these vectors are unit vectors and $\left(\overrightarrow{u}, \overrightarrow{p}\right) \equiv a$, $\left(\overrightarrow{u}, \overrightarrow{q}\right) \equiv b \pmod{2\pi}$. By Chasles' relation, $\left(\overrightarrow{q}, \overrightarrow{p}\right) \equiv a - b \pmod{2\pi}$. We compute $\overrightarrow{q} \cdot \overrightarrow{p}$ in two ways:

$$\overrightarrow{q} \cdot \overrightarrow{p} = \left\|\overrightarrow{q}\right\| \left\|\overrightarrow{p}\right\| \cos(a-b) = \cos(a-b) \quad \text{and} \quad \overrightarrow{q} \cdot \overrightarrow{p} = \cos b \cos a + \sin b \sin a.$$

This proves the first formula. Replacing b by $-b$ and using $\cos(-b) = \cos b$, $\sin(-b) = -\sin b$, we obtain $\cos(a+b)$. Finally $\sin(a+b) = \cos\left(\frac{\pi}{2} - a - b\right) = \cos\left(\left(\frac{\pi}{2} - a\right) - b\right)$, which the first formula expands into $\sin a \cos b + \cos a \sin b$; we deduce $\sin(a-b)$ by replacing b by $-b$. □

Property · Double-angle formulas. For every real number a: $\sin 2a = 2\sin a \cos a$ and $\cos 2a = \cos^2 a - \sin^2 a = 2\cos^2 a - 1 = 1 - 2\sin^2 a$.

Example 1 · An exact value. $\cos\frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{1}{2} \times \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

Product and arguments

Property · Product of two trigonometric forms. For all real numbers θ and θ': $(\cos\theta + i\sin\theta)(\cos\theta' + i\sin\theta') = \cos(\theta+\theta') + i\sin(\theta+\theta')$.

Consequently, for all nonzero complex numbers z, z' and every integer n:

$$\arg(zz') \equiv \arg z + \arg z', \qquad \arg\frac{1}{z} \equiv -\arg z,$$

$$\arg\frac{z}{z'} \equiv \arg z - \arg z', \qquad \arg\left(z^n\right) \equiv n\arg z \pmod{2\pi}.$$

Proof. Expanding gives $(\cos\theta\cos\theta' - \sin\theta\sin\theta') + i(\sin\theta\cos\theta' + \cos\theta\sin\theta')$, which the addition formulas simplify. If $z = r(\cos\theta + i\sin\theta)$ and $z' = r'(\cos\theta' + i\sin\theta')$, then $zz' = rr'(\cos(\theta+\theta') + i\sin(\theta+\theta'))$ with $rr' > 0$: this is a trigonometric form of zz'. The other relations follow: $z \times \frac{1}{z} = 1$ has argument 0, then $\frac{z}{z'} = z \times \frac{1}{z'}$, and an induction for $n \geq 0$, extended to $n < 0$ through the inverse. □

Key point · Interpretation. Multiplying by $\cos\theta + i\sin\theta$ turns through an angle θ about O. The conjecture of the activity of chapter 2 is proved: multiplying by i is a direct quarter turn, multiplying by $1 + i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ is an eighth of a turn followed by an enlargement with ratio $\sqrt{2}$.

Exponential form

Definition · Imaginary exponential. For every real number θ, we write $e^{i\theta} = \cos\theta + i\sin\theta$. Every nonzero complex number can be written $z = re^{i\theta}$ with $r = |z|$ and $\theta \equiv \arg z \pmod{2\pi}$: this is its *exponential form*.

Property · Rules of computation. For all real numbers θ, θ' and every integer n:

$$e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')}, \qquad \frac{1}{e^{i\theta}} = e^{-i\theta} = \overline{e^{i\theta}}, \qquad \frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')}, \qquad \left(e^{i\theta}\right)^n = e^{in\theta}.$$

Values to know: $e^{i0} = 1$, $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{-i\pi/2} = -i$, $\left|e^{i\theta}\right| = 1$.

Property · De Moivre's formula. For every real number θ and every integer n: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$.

Property · Euler's formulas. For every real number θ: $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$.

Proof. $e^{i\theta} + e^{-i\theta} = e^{i\theta} + \overline{e^{i\theta}} = 2\,\mathrm{Re}\left(e^{i\theta}\right) = 2\cos\theta$, and $e^{i\theta} - e^{-i\theta} = 2i\,\mathrm{Im}\left(e^{i\theta}\right) = 2i\sin\theta$. □

Method · Choosing the suitable form. For a sum, the algebraic form; for a product, a quotient or a power, the exponential form. Example: $1 + i\sqrt{3} = 2e^{i\pi/3}$, so

$$\left(1 + i\sqrt{3}\right)^{10} = 2^{10}e^{10i\pi/3} = 1{,}024\,e^{-2i\pi/3} = -512 - 512\sqrt{3}\,i,$$

since $\frac{10\pi}{3} - 4\pi = -\frac{2\pi}{3}$.

Method · Expressing $\cos 3\theta$ in terms of $\cos\theta$: de Moivre. Expand $(\cos\theta + i\sin\theta)^3$ with the binomial formula and identify the real part: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$.

Method · Linearising $\cos^3\theta$: Euler. $\cos^3\theta = \dfrac{\left(e^{i\theta} + e^{-i\theta}\right)^3}{8} = \dfrac{e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta}}{8} = \dfrac{\cos 3\theta + 3\cos\theta}{4}$.

Method · Half angle. To factorise a sum of exponentials, take out the exponential of the mean angle: $e^{ia} + e^{ib} = e^{i\frac{a+b}{2}}\left(e^{i\frac{a-b}{2}} + e^{-i\frac{a-b}{2}}\right) = 2\cos\frac{a-b}{2}\,e^{i\frac{a+b}{2}}$.

Watch out. $e^{i\theta}$ is not a positive real number: its modulus is 1, but it can be negative ($e^{i\pi} = -1$) or non-real. An expression $re^{i\theta}$ is an exponential form only if $r > 0$.